

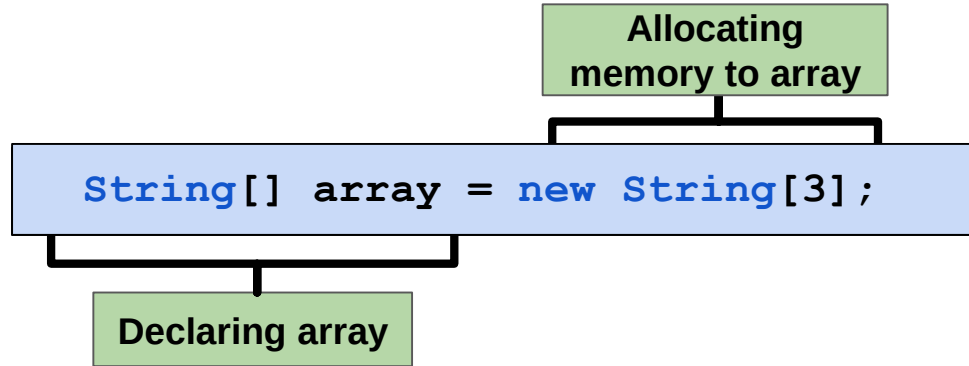
Arrays

What is an array?

- An array is a collection of values of the same data type.
- The length of an array remains constant once it's created.
- Multiple elements are separated using comma when initialized.
- Arrays are considered as objects in java.
- Number of elements of an array is called the length of an array.

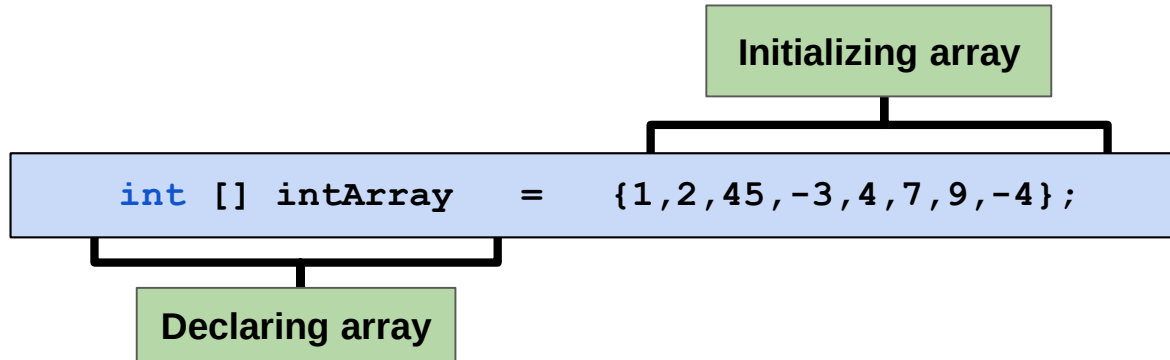
Array Declaration

- “new” Keyword is used to create arrays.
- While initializing an array we declare the type of the elements and the size of the array
- Initially array contains the default values of the specified data type.
- Syntax:



Array Initialization

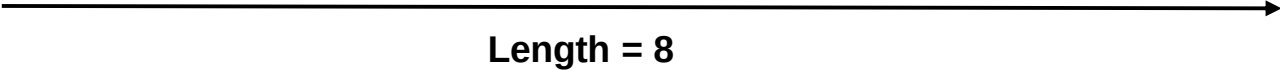
- We can initialize an array at the time of declaration.
- Initializing using an array initializer (a comma-separated list of expressions enclosed in braces)
- Syntax:



Index of an Array

- Each elements of an array holds a position or index.
- The first element in every array has index **zero**.
- The last element has index number (**array.length - 1**)
- Index number is enclosed by square brackets []

Elements	44	3	1	12	-10	38	0	55
Indices	0	1	2	3	4	5	6	7



Length = 8

Array Initialization Example

```
1. class Array {  
2.     public static void main(String[] args) {  
3.         int [] array = new int [] {1,2,3,4}; // [1, 2, 3, 4]  
4.         int [] array2 = {1,2,3,4,7,9}; // [1, 2, 3, 4, 7, 9]  
5.         int [] array3 = new int [3]; // [0, 0, 0]  
6.         array3[0] = 9; // [9, 0, 0]  
7.         array3[1] = 5; // [9, 5, 0]  
8.         array3[2] = 16; // [9, 5, 16]  
9.     }  
10. }
```

We can also initialize an array using loops and taking inputs from the user.

Array Operations

- **Access array elements:**
 - Array elements can be accessed using the array name followed by an index enclosed by square brackets.
 - Syntax: `array_name[index]`
- **Length of an array:**
 - Measures the number of elements in an array
 - **.length** is used to count the number of array elements.

```
String [] courses = {"CSE110", "PHY111", "MAT110", "ENG101"};  
System.out.println(courses[1]); //Accessing element of 1st index  
System.out.println(courses[2]); //Accessing element of 2nd index  
System.out.println(courses.length);
```

```
Welcome to DrJava. Working directory is C:\Users\Desktop  
PHY111  
MAT110  
4
```

Array Operations (Continued)

- **Update array elements:**

- We can change the value of a specific element using the index number.
- Syntax: `array_name[index] = updated_value`

```
// Changing PHY111 to BIO101
```

```
String [] courses = {"CSE110", "PHY111", "MAT110", "ENG101"};  
courses[1] = "BIO101";  
System.out.println(courses[1]);
```

```
Welcome to DrJava. Working directory is C:\Users\Desktop  
BIO101
```

Array Operations (Continued)

- **String representation of an array:**
 - We cannot print array elements directly in Java as it shows the memory location of that array object.
 - **Arrays.toString()** method is used to represent an array.
 - The array elements are separated by comma and enclosed by square brackets [].

```
import java.util.Arrays;  
String [] courses = {"CSE110", "PHY111", "MAT110", "ENG101"};  
System.out.println(courses);  
System.out.println(Arrays.toString(courses));
```

```
Welcome to DrJava. Working directory is C:\Users\Desktop  
[Ljava.lang.String;@379619aa  
[CSE110, PHY111, MAT110, ENG101]
```

Array Iteration

- Array iteration means accessing each element of an array one by one.
- We can use the concept of loops to iterate an array.

Array Iteration - for loop

```
1. class ArrayIteration{
2.     public static void main(String[] args) {
3.         int [] array = new int [] { 44, 3, 1, 12, -10};
4.         for (int i = 0; i<array.length; i++){
5.             System.out.println("Index:"+i+ " Element: "+ array[i]);
6.         }
7.     }
8. }
```

Welcome to DrJava. Working directory is C:\Users\Desktop

> run ArrayIteration

Index:0 Element: 44

Index:1 Element: 3

Index:2 Element: 1

Index:3 Element: 12

Index:4 Element: -10

Array Iteration - while loop

```
1. class ArrayIteration2{
2.     public static void main(String[] args) {
3.         int [] array2 = {1,2,3,4};
4.         int count = 0;
5.         while (count<array2.length){
6.             System.out.println(array2[count]);
7.             count++;
8.         }
9.     }
10. }
```

Welcome to DrJava. Working directory is C:\Users\Desktop

> run ArrayIteration2

1
2
3
4

Now try writing the code for creating an array to store names of six students, where each provide their name as input.

Hint: Initiate a String array of length 6 and use a loop to take input and update each value.

THANK YOU!